

```
import cv2
import numpy as np
import matplotlib.pyplot as plt
from google.colab import files
from google.colab.patches import cv2_imshow # Import cv2_imshow

# Upload the image file
uploaded = files.upload()

# Get the name of the uploaded file
image_filename = list(uploaded.keys())[0]

image = cv2.imread(image_filename)

# Ensure image is loaded correctly, otherwise cv2_imshow will error
if image is None:
    print(f"Error: Could not load image from {image_filename}")
else:
    kernel = np.array([[0, 1, 0],
                       [1, -8, 1],
                       [0, 1, 0]])
    sharpened = cv2.filter2D(image, -1, kernel)

    # Convert BGR to RGB for matplotlib display
    image_rgb = cv2.cvtColor(image, cv2.COLOR_BGR2RGB)
    sharpened_rgb = cv2.cvtColor(sharpened, cv2.COLOR_BGR2RGB)

    plt.figure(figsize=(10, 5))

    plt.subplot(1, 2, 1)
    plt.imshow(image_rgb)
    plt.title('Original Image')
    plt.axis('off')

    plt.subplot(1, 2, 2)
    plt.imshow(sharpened_rgb)
```

```
plt.title('Sharpened Image')  
plt.axis('off')  
  
plt.tight_layout()  
plt.show()
```

[Choose Files](#) Screenshot... 173734.png

Screenshot 2026-08-17 173734.png(image/png) - 196528 bytes, last modified: 8/17/2026 - 100% done

Saving Screenshot 2026-08-17 173734.png to Screenshot 2026-08-17 173734 (1).png

Original Image



Sharpened Image



